Overnight Return Prediction Across Indian Equities

Four-output forecasting framework using daily, intraday and cross-sectional information

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Universe	History	Inputs	Outputs
208 Indian equities	1 June 2020 to 30 June 2026	Daily OHLCV and 1-minute OHLCV	Direction, magnitude and two confidence scores

Research objective: produce leakage-controlled, reproducible forecasts at the close of session T for the next available session open.

1. Executive Summary

Overnight returns are driven by both a common market gap and stock-specific state variables. The modelling design combines recent gap, return, volatility, volume and intraday-path information with cross-sectional state variables observable at the close. The four required outputs are treated as separate statistical questions rather than transformations of a single score.

The final architecture uses a pooled LightGBM classifier for direction, a pooled LightGBM regressor for magnitude, logistic regression for direction correctness, and a separate LightGBM expected-error model for magnitude confidence. Symbol is retained as a categorical effect, allowing information sharing across 208 equities while preserving persistent stock-level heterogeneity.

Output	Selected method	Clean selection evidence	Final test result
Direction	LightGBM classifier; threshold 0.64	Validation direction score 0.2948	Direction score 0.2428; hit rate 59.16%
Magnitude	LightGBM L1 on log1p(|return|)	Validation score 0.2543; rank IC 0.2459	Score 0.2470; rank IC 0.1886
Direction confidence	Logistic correctness model	Selection score 0.2689	Confidence score 0.4019
Magnitude confidence	LightGBM expected-error model	Selection Spearman about 0.30	Error-confidence Spearman 0.2907

The strongest result is the magnitude and confidence stack. Direction remains positive out of sample but is the least settled component: the pooled test score is almost identical to the always-up pooled benchmark, and performance weakens late in the test period. The final submission therefore emphasises honest validation, calibration and reproducibility rather than a single headline score.

2. Data, Target Construction and Validation

The supplied universe contains 208 symbol files. Daily files contain date, open, high, low, close and volume; minute files contain 09:15-15:29 IST bars. Minute features are aggregated one symbol at a time to control memory usage and tolerate incomplete sessions when the required calculations remain finite.

Stage	Rows	Notes
Raw daily panel	301,483	No duplicate symbol-date rows; positive prices
Target-aligned panel	301,275	Final row per symbol removed because T+1 open is unavailable
Final scored rows	295,060	Train 182,495; validation 53,909; test 58,656 after warm-up filtering

For symbol i at session T, the signed target is 100 x (open(i,T+1) / close(i,T) - 1). Exact zeros are assigned direction +1; magnitude is the absolute return. T+1 is the next available trading session for the symbol, not the next calendar day.

All rolling inputs are backward-looking. Target, next-open and future-derived columns are excluded from model matrices. A five-session embargo separates chronological train, validation and test blocks. Validation selection uses models fitted on train only; final test predictions are generated after refitting the primary models on train plus validation.

3. Feature Engineering and Economic Intuition

Family	Representative features	Economic hypothesis
Daily state	Lagged overnight return; 1d/5d/20d returns; 5d/20d volatility; gap	Overnight risk depends on recent trend, gap persistence, uncertainty, liquidity

	mean/std/sign frequency; volume z-score and trend; weekday; calendar gap	attention and time elapsed since the prior close.
Minute-derived	Intraday realised volatility; close-auction return and volume concentration; morning-vs-afternoon return; close-VWAP deviation; illiquidity and path-efficiency measures	The path into the close contains information about unresolved order flow, closing pressure, liquidity and information likely to carry overnight.
Cross-sectional	Return rank and z-score; market breadth; cross-sectional dispersion; aggregate universe volatility; beta to the universe	A stock's overnight move is conditional on the common market state and on its relative position within the universe.

The final direction specification uses 25 numeric features plus symbol. The magnitude model uses a wider causal feature set selected programmatically after excluding targets and suspicious forward-looking names. Same-day cross-sectional ranks, breadth and dispersion are computed using information available by the close of T and therefore remain inside the admissible information set.

4. Model Architecture and Iteration

Problem	Target / transformation	Final model
Direction	1 if overnight return >= 0, else 0	LightGBM binary classifier; 31 leaves; learning rate 0.05; 1,000 trees; threshold 0.64
Magnitude	log1p(abs overnight return)	LightGBM L1 regressor; 31 leaves; learning rate 0.03; validation-selected 84 trees
Direction confidence	1 if emitted direction is realised	Logistic regression, C=0.25, trained on validation-stage out-of-sample correctness labels
Magnitude confidence	log1p(abs magnitude forecast error)	LightGBM L1 expected-error model; confidence is inverse percentile of expected error

Pooling avoids fitting 208 unstable symbol-specific models and allows information sharing across the panel. Confidence is modelled separately because a class probability estimates P(return > 0), not the probability that the thresholded emitted action is correct; magnitude confidence similarly estimates expected forecast error rather than rescaling the magnitude output.

Experiment	Observed outcome	Decision
Logistic direction baseline	Useful ranking but strong majority-up tendency	Retained as benchmark only
Direction V3 feature set	Reconciled validation score 0.3112 versus V1 0.2948	Documented as next improvement; not promoted after final stack freeze
Magnitude Huber objective	Lower RMSE / higher R2 but lower primary magnitude score	Rejected
Large-tree L1 magnitude	Only 0.00008 score improvement while rank IC and R2 weakened	Rejected as validation-score chasing
Shallow LightGBM direction confidence	Lower confidence score and Brier skill	Rejected; logistic retained

5. Final Results

Metric	Train	Validation	Test
Direction score	0.800860	0.294817	0.242845
Hit rate	0.856966	0.650893	0.591636
Magnitude score	0.361962	0.252219	0.246985
Magnitude rank IC	0.317327	0.246034	0.188596
Direction-confidence score	0.860855	0.456954*	0.401853
Magnitude-confidence Spearman	0.303998	0.425866*	0.290722

** Validation confidence metrics are in-sample with respect to the final confidence models because those models are trained on validation correctness and magnitude-error labels. Test confidence metrics are the clean out-of-sample assessment.*

The test direction score is positive but almost identical to the always-up pooled direction score of 0.243361. Test hit rate is 59.16% while the actual-up fraction is 66.75%, so hit rate alone is not evidence of incremental skill. Magnitude performance is materially more stable from validation to test, declining only from 0.2522 to 0.2470.

Target alignment was manually verified on sampled symbols and dates: the direction match rate was 100%, and the maximum absolute return difference versus recomputation from raw daily files was 2.22e-16.

6. Breadth and Cross-sectional Behaviour

Breadth statistic	Test value
Eligible symbols	208
Symbols with hit rate above 50%	95.67%
Symbols beating own always-up hit rate	4.33%
Symbols with positive direction score	95.67%
Symbols beating own always-up direction score	51.44%
Median symbol hit rate	59.22%
Median symbol direction score	0.2513

Positive weighted-direction performance is broad, but incremental classification skill over each stock's own upward base rate is limited. This distinction supports interpreting the system as a common-factor-aware overnight forecaster with modest stock-selection content, rather than a pure cross-sectional alpha model.

Top test symbols by direction score	Score	Bottom test symbols by direction score	Score
SUZLON	0.5496	HINDUNILVR	-0.0830
ADANIPOWER	0.5436	MPHASIS	-0.0784
RECLTD	0.4979	TCS	-0.0625
FORCEMOT	0.4755	RELIANCE	-0.0548
COCHINSHIP	0.4449	HCLTECH	-0.0223

7. Stability Through Time

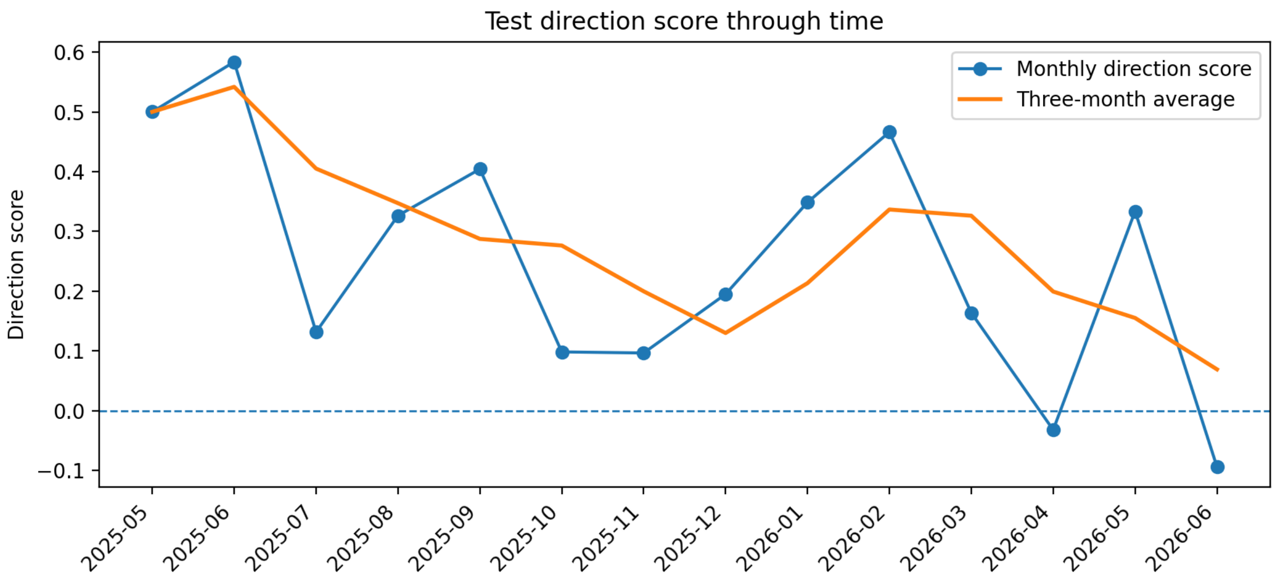

Figure 1. Monthly test direction score and three-month average.

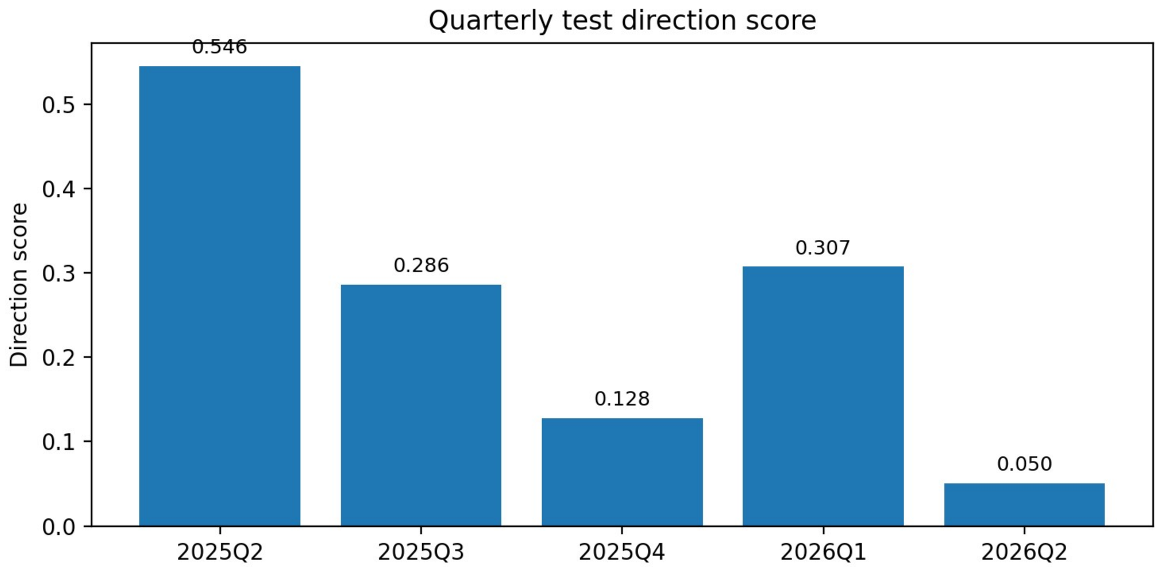

Figure 2. Quarterly test direction score.

Performance is not stationary. Direction score declined from 0.5457 in 2025Q2 to 0.0503 in 2026Q2. April 2026 (-0.0320) and June 2026 (-0.0940) were negative months. The late-sample deterioration argues for expanding-window retraining, regime diagnostics and a more explicit residual objective before deployment.

8. Pooled versus Residual and Transaction Costs

The architecture is deliberately designed to capture the pooled overnight gap while adding stock-relative information. Market breadth and aggregate volatility are among the most informative state variables. The residual scope in statistics.csv removes the daily universe mean from realised returns, providing a direct check on how much of the forecast survives after the common overnight factor is stripped out.

Metric	**Pooled test**	**Residual test**	**What survives residualisation**
Direction score	0.242845	0.077855	Positive, but much smaller
Directional return (%)	0.128071	0.029707	Most pooled directional return is common-factor
Hit rate	0.591636	0.522010	Falls close to 50%
Magnitude score	0.246985	0.242670	Largely preserved
Magnitude rank IC	0.188596	0.104523	Positive stock-relative ranking remains
Direction-confidence score	0.401853	0.090856	Most direction-confidence signal is pooled
Magnitude-confidence score	0.290722	0.321414	Reliability signal remains strong

The residual comparison materially changes the interpretation. Direction score falls from 0.242845 pooled to 0.077855 residual, and hit rate falls from 59.16% to 52.20%, showing that much of the direction result is tied to the common overnight factor. Magnitude is more robust: its score declines only from 0.246985 to 0.242670, while residual rank IC remains positive at 0.104523. Magnitude-confidence also remains strong at 0.321414 residual versus 0.290722 pooled. This suggests the most defensible stock-specific information in the final stack lies in forecasting move size and forecast reliability rather than in pure residual direction.

Directional return is a frictionless diagnostic, not implementable P&L. Trading near the close and reversing or rebalancing near the next open introduces auction impact, bid-ask spread, slippage, taxes and borrow constraints. A deployable strategy should trade sparsely using both confidence outputs, predicted magnitude and liquidity, and should report turnover and net performance across realistic cost bands.

9. Reproducibility and Production Design

- One config-driven entry point rebuilds features, trains all four models, scores the required splits and writes predictions.csv, actuals.csv and the 123-row statistics.csv.
- All paths are relative; seeds and package requirements are fixed; no hard-coded local paths are required.
- The slow minute stage is processed symbol-by-symbol and supports a validated cache; cache-free execution remains available.
- Runtime assertions enforce key alignment, duplicate absence, finite values, valid direction classes, confidence ranges and exact statistics completeness.
- The final cache-hit pipeline completed successfully in approximately 55 seconds on the development machine.

Reproduction command: python code/main.py --config code/config.yaml

Production check	Final result
Predictions / actuals key alignment	PASS
Output rows	295,060 each
Statistics completeness	123 rows
Direction benchmark	0.294817 reproduced
Magnitude benchmark	0.254340; rank IC 0.245931; 84 trees
Direction-confidence benchmark	0.268893 reproduced
Magnitude-confidence benchmark	Approximately 0.30 Spearman reproduced

10. Limitations, Prioritisation and Next Steps

Limitation / dropped item	Reason and implication
Direction model freeze despite V3 result	The final confidence stack and outputs were already frozen. Promoting V3 would require retraining confidence models and regenerating all outputs on identical rows.
Confidence validation not cross-fitted	Final validation confidence values are in-sample; test values remain clean. Cross-fitting is the first calibration improvement.
No walk-forward retraining	A single chronological split was prioritised for clarity and reproducibility; late-test deterioration remains a material risk.
No point-in-time sector or fundamental data	Avoided sourcing risk and hindsight-biased classifications.
No net-cost strategy backtest	The assignment evaluates forecasts; a portfolio, execution and risk layer is a separate mandate.

Highest-value next steps are to (i) rerun the full four-output stack with V3 direction features, retaining row-level V1/V3 predictions; (ii) cross-fit confidence models; (iii) build a residual-first target relative to the universe mean; (iv) use expanding-window retraining; and (v) evaluate a sparse, cost-aware decision layer using both confidence outputs, magnitude and liquidity.

11. Conclusion

The project delivers four leakage-controlled forecasts from daily, minute and cross-sectional information for 208 Indian equities. The strongest empirical result is the magnitude and confidence stack: expected magnitude improves on the baseline, ranking remains positive out of sample, and magnitude performance remains largely intact after removing the daily universe mean. Direction also produces a positive test score, but the residual score is materially smaller and performance weakens late in the sample.

The final submission therefore presents the system as a reproducible overnight forecasting framework with useful magnitude and reliability information, but only modest incremental residual-direction content. The research trail, rejected experiments, limitations and next steps are stated explicitly rather than hidden behind a single aggregate score.